Evolution of Transport-Layer Functionality

(TCP Diversity and the QUIC Protocol)

1. The Diversity of Modern TCP

While "Classic" TCP (Tahoe and Reno) provided the foundation, three decades of experience have led to a rich ecosystem of TCP variants designed for specific environments.

- **General Purpose:** TCP CUBIC (Linux default), BIC, and CTCP (Windows).
- **High-Speed & Internal:** BBR (Google), DCTCP (Data Centers).
- **Specialized Variants:**
  - Wireless links (handling non-congestion loss).
  - High-bandwidth paths with large RTTs (Long Fat Pipes).
  - Paths with significant packet re-ordering.
  - Multi-path TCP (using multiple paths in parallel).

A Unified TCP?

Modern transport is so diverse that the term "TCP" often refers more to the **segment format** and the general requirement of **fairness** rather than a single fixed algorithm.

2. QUIC: Quick UDP Internet Connections

When standard UDP or TCP doesn't fit, developers can "roll their own" at the application layer. **QUIC** is a revolutionary application-layer protocol designed to improve secure HTTP performance.

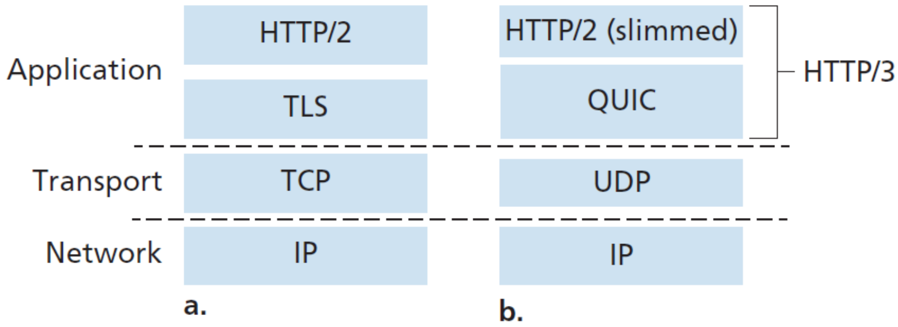

Figure 1: Evolution of the Protocol Stack: (a) Traditional Secure HTTP vs (b) QUIC-based HTTP/3

2.1 Key Features of QUIC

2.1.1 1. Connection-Oriented and Secure

Unlike the sequential handshakes of TCP and TLS, QUIC combines them.

- **Reduced Latency:** Establishes connection state and encryption keys simultaneously, reducing the RTTs required for a secure session.
- **Encryption by Default:** All QUIC packets are encrypted at the transport level.

2.1.2 2. Streams and Multiplexing

QUIC allows multiple application-level streams to be multiplexed through a single connection.

- **Abstractions:** Each connection has a unique **Connection ID**, and each stream within it has a **Stream ID**.
- **Efficiency:** Data from different streams can be packed into a single UDP segment.

2.1.3 3. Solving Head-of-Line (HOL) Blocking

This is the most significant architectural improvement over TCP-based HTTP/2.

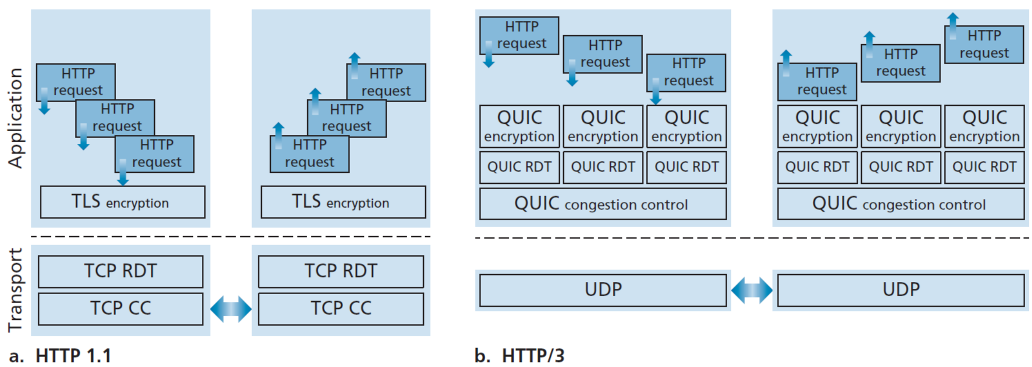

Figure 2: Reliable Data Transfer: (a) TCP HOL Blocking vs (b) QUIC Independent Streams

Per-Stream Reliability

In TCP, if a packet is lost, all subsequent data (even for unrelated streams) is held up until retransmission.
In **QUIC**, reliability is on a **per-stream basis**. A lost UDP segment only impacts the streams it carried; other streams continue to be delivered to the application without delay.

3. Congestion Control in QUIC

QUIC's congestion control is not a radical departure but a refinement. It is based on **TCP NewReno** and allows for "application-update timescale" changes, meaning it can be updated and improved much faster than kernel-level TCP.

4. Summary: The Culmination of Transport Study

The Transport Layer Landscape

- **UDP:** Simple, no-frills multiplexing.
- **TCP:** Complex, reliable, congestion-controlled.
- **QUIC:** The hybrid future—reliability and security over UDP, eliminating traditional TCP bottlenecks like HOL blocking.